

THE FORGOTTEN FIELD AGENT

// CHALLENGE WRITUP – OSINT TRACK

CATEGORY	DIFFICULTY	POINTS	STAGES	AUTHOR
OSINT	HARD	500	5	NSC-TEAM

EYES ONLY

00 // CHALLENGE DESCRIPTION

An operative known only as **R4VENOUS** went dark in late 2021. No formal exit. No forwarding address. Just silence. A conference flyer was recovered from a dead drop in Casablanca. Intel suggests he didn't disappear – he left a trail. Intentionally.

Your mission: find what he left behind.

⚠ The flag is not in the document. It never was. | Flag format: **NSC{...}**

01 // ATTACK CHAIN OVERVIEW

1	NorthSecConf_2026_Program.pdf i Author/Comment field reveals github.com/ravenous	– exiftool metadata
2	github.com/ravenous i README hints at dead blog → Wayback Machine	– profile README
3	web.archive.org → ravenous-notes.netlify.app i Blog post has drop_audio.wav + keyword "Argos-2"	– blog post
4	github.com/ravenous/drop_audio i config/signal_params.json has Argos-2 decoding table	– OSINT on ravenous repos
5	Spectrogram → DTMF sequence → Argos-2 decode	– flag

02 // STAGE 1 – PDF METADATA

PHASE 01

EXTRACT GITHUB URL VIA EXIFTOOL

Players are given **NorthSecConf_2026_Program.pdf** – a realistic 12-page conference program for NorthSec Research Conference 2026, held in Tangier, Morocco. **R4VENOUS** is listed as a speaker on Day 1 at 13:30 in Room A. The document contains no visible links, URLs, or hints.

The first move is to inspect the file metadata:

```
$ exiftool NorthSecConf_2026_Program.pdf

File Name       : NorthSecConf_2026_Program.pdf
Author          : ravenous
Keywords        : medsec2021, network-security, ravenous
Comment        : Speaker profile: github.com/ravenous
Creator         : LibreOffice 7.4
PDF Producer    : NorthSec Research Collective
```

```
// FINDING
GitHub profile URL extracted from Comment metadata field:
github.com/ravenous
```

DECOYS IN THIS STAGE

The PDF contains several deliberate rabbit holes to mislead players:

DECOY	LOCATION	WHY IT'S A DEAD END
iv3n0sus.research@proton.me	Speaker bio (p.5)	Inbox does not exist – no response
@iv3n0sus (Telegram)	Contact table (p.12)	Different spelling – account does not exist
archive.org/details/northsec2021-proceedings	Hidden white-on-white text (p.8)	URL returns 404
Tokyo timezone timestamp	PDF creation metadata	Wrong geolocation – misleads EXIP hunters

03 // STAGE 2 – GITHUB PROFILE

PHASE 02

FIND THE DEAD BLOG URL VIA PROFILE README

Players visit **github.com/ravenous**. The profile has 5 public repositories with realistic content and historical commit dates. The profile README contains:

```
## ravenous
Security researcher. Occasional conference speaker. Full-time skeptic.

Personal notes were at ravenous-notes.netlify.app – site is down, migrating
If you need the old content:
> The Wayback Machine usually has it.
```

REPOSITORIES (DECOYS + ONE KEY)

REPO	ROLE
ravenous/ravenous	Profile README – contains Wayback Machine hint
netflow-tools	Decoy – realistic Python NetFlow parser scripts
anomaly-notes	Decoy – research notes on anomaly detection papers
conf-slides-archive	Decoy – "missing" slide files, dead end
scratchpad	Decoy – throwaway scripts including b64check.py
drop_audio	Key repo – contains Argos-2 decoding table in config/

```
// FINDING
Dead blog URL recovered: ravenous-notes.netlify.app
Players must search the Wayback Machine for an archived snapshot.
```

04 // STAGE 3 – ARCHIVED BLOG & AUDIO FILE

PHASE 03

LOCATE DROP_AUDIO.WAV ON THE WAYBACK MACHINE SNAPSHOT

Players search the Wayback Machine for the dead blog:

```
https://web.archive.org/web/7/ravenous-notes.netlify.app
```

A single snapshot exists from **April 13, 2026**. The archived blog has 3 posts. The MedSec 2021 post contains an embedded audio player and the following line:

```
"Recovered a field audio log from the trip – leaving it here.
Format follows the old Argos-2 signaling spec – if you know, you know."
```

Direct snapshot URL:

```
https://web.archive.org/web/20260413083501/https://ravenous-notes.netlify.app/
```

```
// FINDING
Audio file drop_audio.wav (8.82s, 44100Hz mono) downloaded.
Keyword extracted: Argos-2 – players must now find what this means.
```

05 // STAGE 4 – THE DROP_AUDIO REPO

PHASE 04

FIND CONFIG/SIGNAL_PARAMS.JSON BURIED PAST DECOY FOLDERS

Players either find the **drop_audio** repo by browsing ravenous's GitHub profile, or by searching "ravenous drop_audio" or "Argos-2 DTMF". The repo presents itself as a legitimate audio signal processing project.

The repo structure contains **5 convincing decoy folders** before players find the key file:

```
drop_audio/
├── src/           – decoy: audio processing source code
├── tests/        – decoy: unit test files
├── docs/         – decoy: fake documentation
├── examples/     – decoy: sample wav files
├── scripts/      – decoy: helper bash scripts
├── config/
└── signal_params.json – THE KEY
```

config/signal_params.json contains the Argos-2 decoding table:

```
{
  "scheme": "argos2",
  "version": "2.1.4",
  "encoding": "19-DTMF",
  "charset": {
    "letters": "[3-keypad-position]",
    "digits": "direct",
    "special": { "[": "A", "]": "B", "_": "C" }
  },
  "keypad": {
    "1": ["A","B","C"], "2": ["D","E","F"],
    "4": ["G","H","I"], "5": ["J","K","L"],
    "6": ["M","N","O"], "7": ["P","Q","R","S"],
    "8": ["T","U","V"], "9": ["W","X","Y","Z"]
  }
}
```

```
// FINDING
Argos-2 encoding table recovered from config/signal_params.json.
Players now have everything needed to decode the audio.
```

06 // STAGE 5 – DTMF DECODE → FLAG

PHASE 05

SPECTROGRAM ANALYSIS + ARGOS-2 DECODE

STEP 1 – OPEN IN AUDACITY

Players open **drop_audio.wav** in Audacity or any spectrogram tool. Switch to spectrogram view: track dropdown → **Spectrogram**. Two simultaneous frequency bands are visible for each tone – these are the DTMF pairs. Ambient noise is present but does not mask the tones.

STEP 2 – READ THE RAW DTMF SEQUENCE

```
# Raw tone sequence read from spectrogram
6 2 | 7 4 | 2 3 | A | A 1 | B | C | B 3 | B | B 2 | 2 3 | 4 2 | C | A 3 | 7 3 | 4 | 2 4 | 5
```

STEP 3 – APPLY ARGOS-2 TABLE

DTMF Keys	Role	Character	Running Flag
6, 2	Key 6 pos 2 → N	N	N
7, 4	Key 7 pos 4 → S	S	NS
2, 3	Key 2 pos 3 → C	C	NSC
A	Special → (	(	NSC(
4, 1	Key 4 pos 1 → G	G	NSC(G
0	Digit → 0	0	NSC(00
C	Special → _	_	NSC(00_
0, 1	Key 0 pos 1 → T	T	NSC(00_T
0	Digit → 0	0	NSC(00_T0
0, 2	Key 0 pos 2 → U	U	NSC(00_T0U
2, 3	Key 2 pos 3 → C	C	NSC(00_T0UC
4, 2	Key 4 pos 2 → H	H	NSC(00_T0UCH
C	Special → _	_	NSC(00_T0UCH_
4, 1	Key 4 pos 1 → G	G	NSC(00_T0UCH_G
7, 3	Key 7 pos 3 → R	R	NSC(00_T0UCH_Gr
4	Digit → 4	4	NSC(00_T0UCH_Gr44
7, 4	Key 7 pos 4 → S	S	NSC(00_T0UCH_Gr4s4s
5	Digit → 5	5	NSC(00_T0UCH_Gr4s4s5
0	Special →)	)	NSC(00_T0UCH_Gr4s4s5)

RECOVERED PAYLOAD

NSC{GO_TOUCH_GR4S5}

07 // TOOLS USED BY SOLVERS

STAGE	TOOL	PURPOSE
1	exiftool	Extract PDF metadata fields
2	Browser / GitHub	Browse ravenous profile and repos
3	web.archive.org	Access archived blog snapshot
4	GitHub repo browser	Dig through drop_audio repo to find config/
5	Audacity / Sonic Visualiser / multimon-ng	Spectrogram + DTMF detection
5	Manual decode + Argos-2 table	Convert DTMF sequence to flag

08 // ANTI-LLM DESIGN NOTES

MECHANISM	WHY IT BLOCKS AI
Flag in PDF metadata	LLMs read document text, not binary metadata
Dead live URL (Wayback only)	Cannot browse live-or-dead archive snapshots
Flag in audio (DTMF)	Cannot perform spectral analysis on raw WAV
Custom Argos-2 scheme	Not a known encoding – hallucination trap
Decoding table buried in repo	Requires navigating past 5 decoy folders
Ambient noise in audio	Masks tones from naive frequency detection
4 PDF decoys	Multiple false trails cause hallucination paths

09 // CHALLENGE ASSETS REFERENCE

FILE	PURPOSE	STATUS
NorthSecConf_2026_Program.pdf	Starting file given to players	✓ Ready
drop_audio.wav	DTMF audio with flag encoded	✓ Ready
generate_dtmf.py	Regenerate audio from any flag	✓ Ready
setup_ravenous	Auto-create all GitHub repos	✓ Ready
ravenous_blog.html	Blog with audio player	✓ Archived
challenge.html	CTF platform challenge page	✓ Ready
config/signal_params.json	Argos-2 table in drop_audio repo	✓ Live on GitHub

10 // PRE-EVENT CHECKLIST

TASK	PRIORITY	NOTES
Verify PDF exiftool output	🔴 Critical	github.com/ravenous must appear in Comment field
Confirm Wayback snapshot loads	🔴 Critical	web.archive.org/web/20260413003501/https://ravenous-notes.netlify.app/
Confirm audio plays in snapshot	🔴 Critical	Audio embedded as base64 in HTML – should work offline
Confirm drop_audio repo is public	🔴 Critical	config/signal_params.json must be accessible
Upload PDF to CTF platform	🔴 Critical	Only file given to players
Set flag on CTF platform	🔴 Critical	NSC{00_Touch_Gr4s5}
Full chain dry run (incognito)	🟡 Important	Time it – should be 60-120 min for expert players
Verify all decoys are dead ends	🟡 Important	No Proton inbox, no Telegram account
Test audio in Audacity	🟡 Important	Confirm tones visible in spectrogram view
Add profile picture to @ravenous	🟢 Optional	Makes account look less freshly created